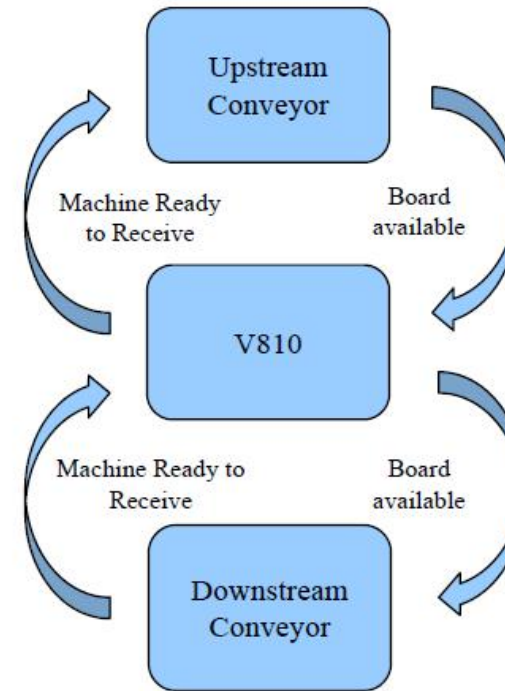
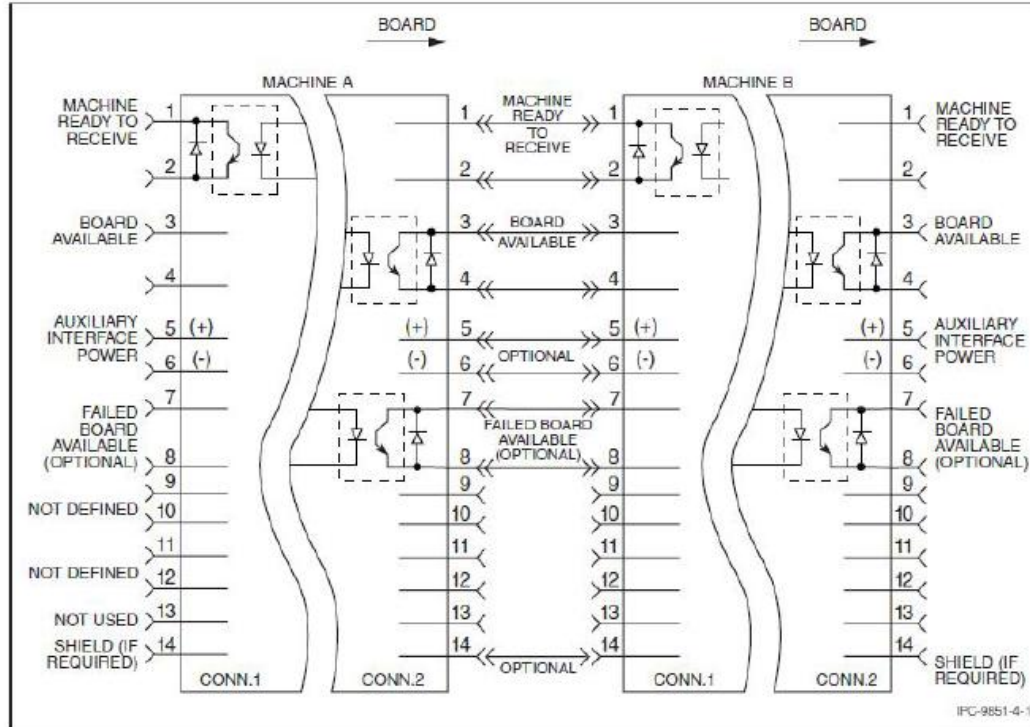


www.vitrox.com

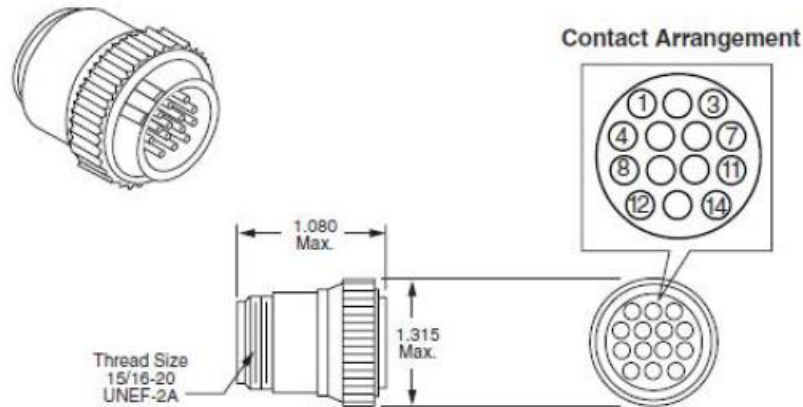
V810i S2EX Automation





- The SMEMA machine interface standards were developed to facilitate the interface of equipment used in the manufacture of surface-mounted printed circuit boards.
- To sequence boards properly from machine-to-machine, the “Board Available” and “Machine Ready” signal lines will be used and “Board Pass/Fail” signal line is optional.

- V810i is complete with SMEMA ready feature where allow V810i communicate with upstream and downstream conveyor to transfer SMT PCBA board in, for inspection and unload, after inspection complete.
- SMEMA connection allocated near Left Outer barrier mainly for board load into V810i machine and SMEMA allocated near Right Outer barrier for unload purpose after inspection complete.
- V810i flexible to run production with or without SMEMA (Manual Load), depend on shop floor plan
 - “Flow Through”, Left In, Right Out
 - “Flow Through”, Right In, Left Out
 - “Pass Back”, Left In, Left Out
 - “Pass Back”, Right In, Right Out
- Machine ready with Good Panel Output (GPO) and Untested Panel Output (UPO) feature to send board to desire downstream loader.
- GPO define as no true/false call rate board after inspection complete.
- UPO define as inspected board with true/false call rate, alignment failed or skip board.



Connector/ Cable	Function	Description
Pair 1-2	Machine Ready to Receive	Machine is ready to receive next board.
Pair 3-4	Board Available	Machine has a good board ready to send. All boards are considered to be “good” if the Board Fail option is not being used.
Pair 5-6	Auxiliary Interface Power (optional)	Available; user to document purpose and operating parameters.
Pair 7-8	Failed Board Available (Optional)	Default (no connection or, if used, contacts are Open) is that the incoming board is good and suitable for use. Optional use is to provide closed contacts when it has been determined that the board should stop transfer or be diverted. In such cases, these contacts shall be closed in lieu of (and not in addition to) the normal Good Board Available contacts.

- V810 SMEMA set up is configured according to IPC SMEMA 9851 Standard.
- With upstream communication, V810 will index “Machine Ready to Receive” to upstream conveyor. Conveyor will index “Board Available” signal to V810 to load in the panel.
- For downstream communication, conveyor will index “Machine Ready to Receive” to V810. V810 will index “Board Available” signal to downstream conveyor to unload panel.
- Upstream and downstream are using exactly the same hand-shaking method for board transfer.



Definition

Automated Board Inspection

Signal	Description of Compliant Interface Connections
TRO	Transfer Request Output
TRI	Transfer Request Input
SRO	Send Request Output
SRI	Send Request Input
GPO	Good panel Output
UPO	Untested Panel Output

v810 Options - Production Options - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Tools Actions Window Help

Software Options Production Options Results Processing Customer Options

General Serial Number Handling Barcode Reader False Call Triggering

Production Options

- ☒ Eject panel if alignment fails
 - ☐ Create result if alignment fails
- ☐ Demo mode
- ☐ Bypass mode
 - ☐ Enable serial number and recipe control during bypass mode
- ☐ Create results during bypass mode
- ☐ Semi-Automated mode
- ☒ Enable system light tower buzzer
- ☐ Always Display Variation Name In Repair Station
- ☐ Display Production Panel Loading and Unloading Time
- ☐ Enable X-out detection
 - ☒ Use Alignment failure as x-out Detection. (Must use individual alignment board Method)
 - ☐ Number of failed joints is greater than % of total number of inspected joints on each board

Confirmation and Adjustment mode:

Sampling Options

- ☐ Panel sampling
 - ☒ Test 1 in every panels OR ☐ Skip 1 in every panels

Panel Handling

- ☒ Use SMEMA communications
 - ☐ Ignore PIP sensor during board loading
 - ☒ Send "Panel From XrayTester Available signal" to downstream after receive "Downstream Waiting Accept Panel Signal" during unloading
 - ☐ Remain Outer Barrier Open For Loading After Unloading Panel (Only Applicable to Pass Back Loading Mode)
 - ☒ Open Outer Barrier Immediately After Scanning Complete
 - ☐ Check Downstream Sensor Clear Signal During Board Unloading

Loading mode:

Flow direction:

Repair Images Save Options

- ☐ Save focus confirmation repair data
 - * Checking this box could increase the number of repair images saved by a factor of 5
- ☐ Limit number of repair images saved
 - Max number of repair images to save
- Repair Image Type ☒ JPG ☐ PNG

Script Setup

- ☐ Run program before panel inspection
 - Path:
- ☐ Run program after panel inspection
 - Path:
- ☐ Run program for skipped panel

☒ Panel Based ☐ Board Based

Pre Inspection Customize Input Key

Sequence No.	Parameter
☐	eqpCode
☐	projectName
☐	userName
☐	panelWidthInNanometer

Pre Inspection Customize Output Key

Sequence No.	Parameter
☐	warning
☐	error
☐	panelSerialNumber

Status:

Panel Handling

☒ Use SMEMA communications

☐ Ignore PIP sensor during board loading

☒ Send "Panel From XrayTester Available signal" to downstream after receive "Downstream Waiting Accept Panel Signal" during unloading

☐ Remain Outer Barrier Open For Loading After Unloading Panel (Only Applicable to Pass Back Loading Mode)

☒ Open Outer Barrier Immediately After Scanning Complete

☐ Check Downstream Sensor Clear Signal During Board Unloading

Loading mode: Flow Thru...

Flow direction: Flow Through
Pass Back

Panel Handling

☐ Use SMEMA communications

☐ Ignore PIP sensor during board loading

☒ Send "Panel From XrayTester Available signal" to downstream after receive "Downstream Waiting Accept Panel Signal" during unloading

☐ Remain Outer Barrier Open For Loading After Unloading Panel (Only Applicable to Pass Back Loading Mode)

☒ Open Outer Barrier Immediately After Scanning Complete

☐ Check Downstream Sensor Clear Signal During Board Unloading

Loading mode: Flow Thru...

Flow direction: Left To Right
Left To Right
Right To Left

- User can self define how to send board in/out with V810i machine, with or without SMEMA (Manual Load), depend on shop floor plan
 - "Flow Through", Left In, Right Out
 - "Flow Through", Right In, Left Out
 - "Pass Back", Left In, Left Out
 - "Pass Back", Right In, Right Out
- Once SMEMA communication enable, few options can be define as image shown.





V810i GUI -SMEMA tab

Automated Board Inspection

File

Edit

Service

Tools

Actions

Window

Help

Display Measurements in: Inches

(KV : 0.0 uA : 0.0)

Main

Recipes

Production

Options

Services

Virtual Live

Confirmation, Diagnostics, & Adjustments

Subsystem Status & Control

X-ray Safety Test

Panel Handling

Panel Positioner

X-ray Source

X-ray Camera

Optical Camera

Ethernet-Based Light Engine

Light Tower

SMEMA

Interlocks

Motion Control

Digital IO

Connected Processors

Hardware Simulation Control

Initialize

SMEMA

TROSystem Ready to Acceptis FalseTrue

TRIDownstream System Ready to Acceptis False

SROPanel From System Availableis FalseTrue

SRIPanel From Upstream System Availableis False

Upstream Machine

v810

Downstream Machine

TRO Ready To Accept

SRI Board Available

TRI Ready To Accept

SRO Board Available

GPO

UPO

Test Results

Set GPO to Passed

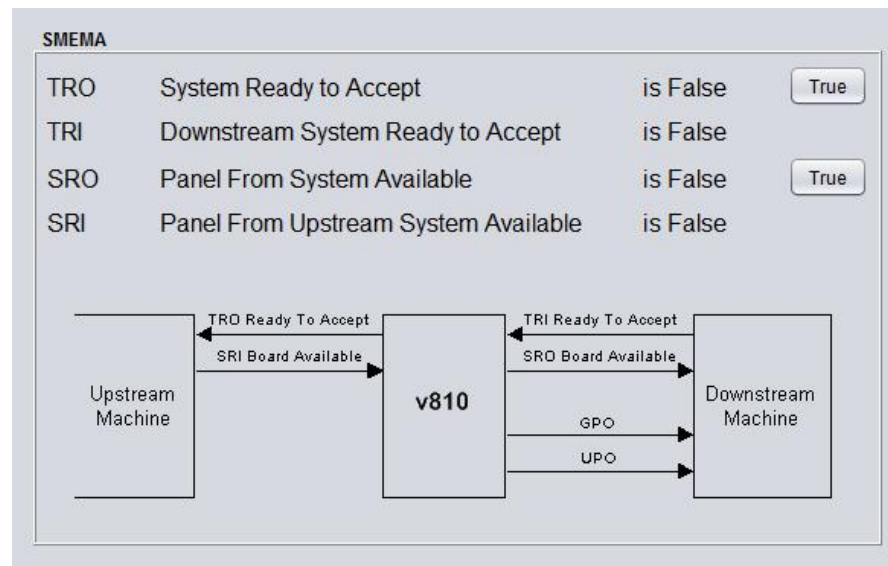
Set GPO to Failed

Set UPO to Untested

Test Result	GPO	UPO
Untested	False	True
Passed	True	False
Failed	False	False

ABI V810i GUI -SMEMA tab

Automated Board Inspection



1. In the SMEMA panel, text messages indicate the status of upstream and downstream devices.
2. Click **True** to change the status of these devices to true. (Changing the status of the upstream and downstream devices to true sends a signal to test for communication.)
3. Check your upstream and/or downstream device to look for an indicator of this signal. If there is a signal, communication is working correctly.
4. Click **False** to change the status of these devices to false. No signals will be sent when the status is false.

ABI V810i GUI -SMEMA tab

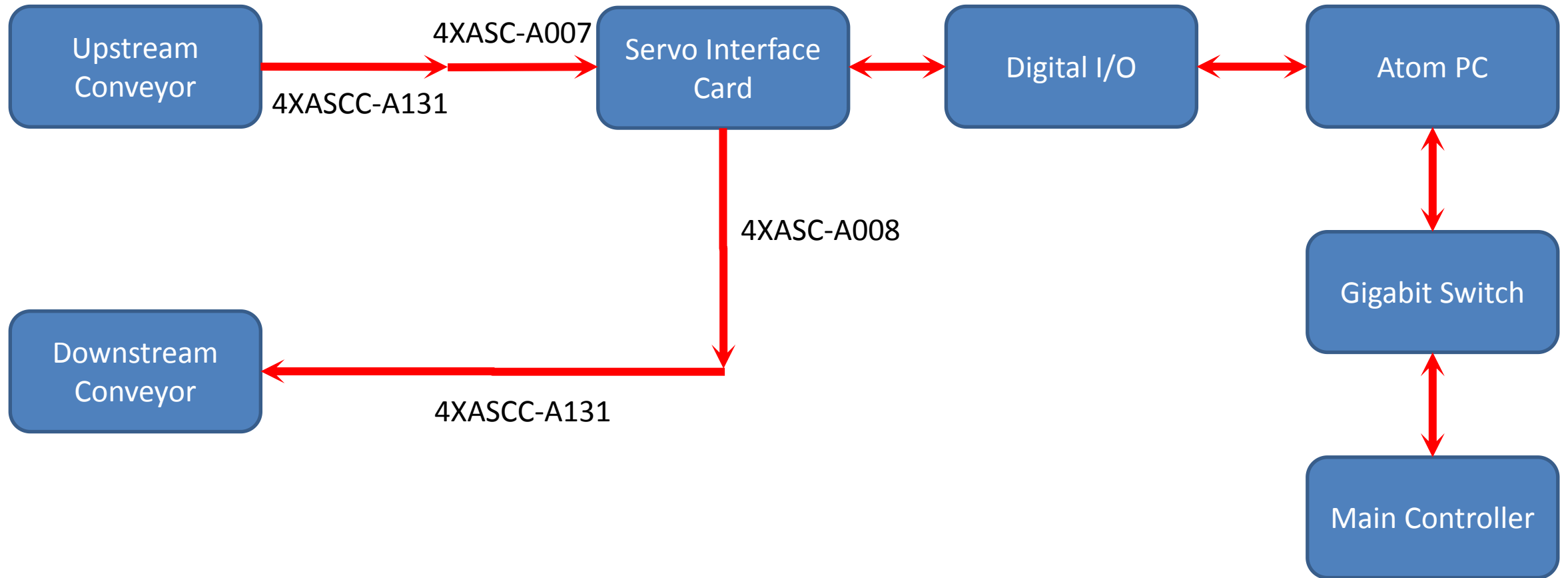
Automated Board Inspection

Test Results

☐ Set GPO to Passed ☐ Set GPO to Failed ☐ Set UPO to Untested

Test Result	GPO	UPO
Untested	False	True
Passed	True	False
Failed	False	False

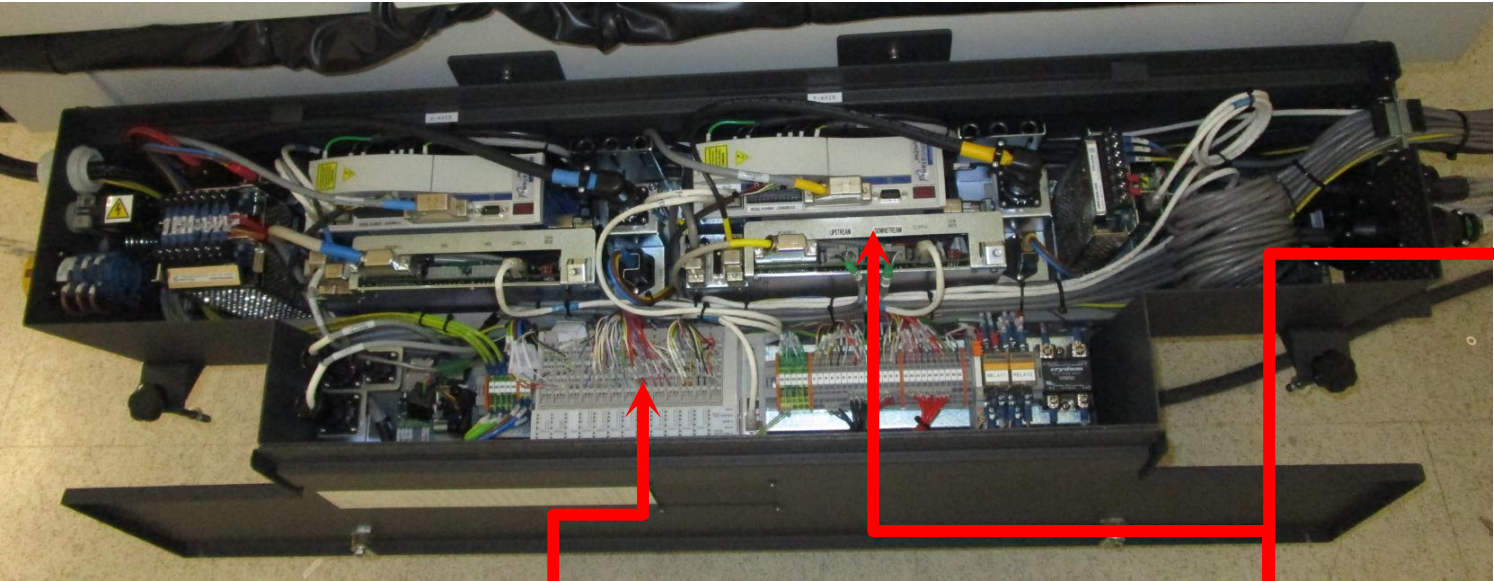
1. In the Test Results panel, there is a read-only table that shows the test results of the most recent panel tested. There are also option buttons available that allow you to send messages to your downstream device.
2. Click **Set GPO to Passed** to send a signal that a board has passed.
3. Click **Set GPO to Failed** to send a signal that a board has failed.
4. Click **Set UPO to Untested** to send a signal that a board was not tested





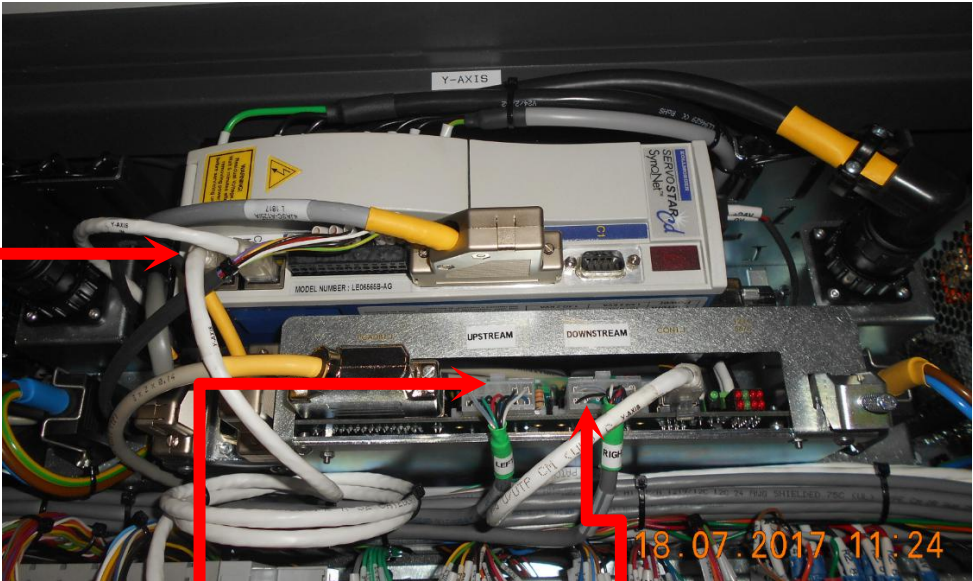
SMEMA Hardware Overview - Y axis Kollmorgen Assembly Kits

Automated Board Inspection



Upstream

Y axis Kollmorgen Assembly Kits



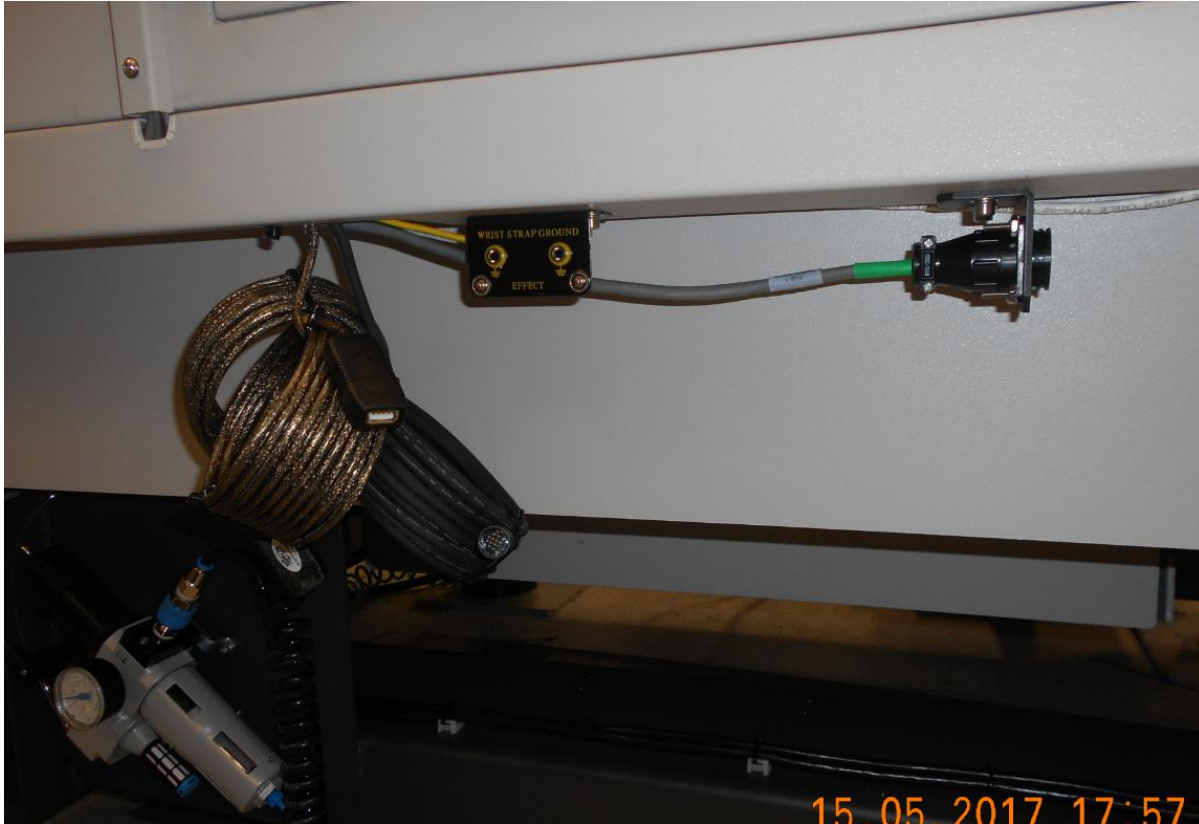
Upstream

Downstream



Automated Board Inspection

SMEMA Hardware Overview - SMEMA Connector (Female)



SMEMA In, Upstream

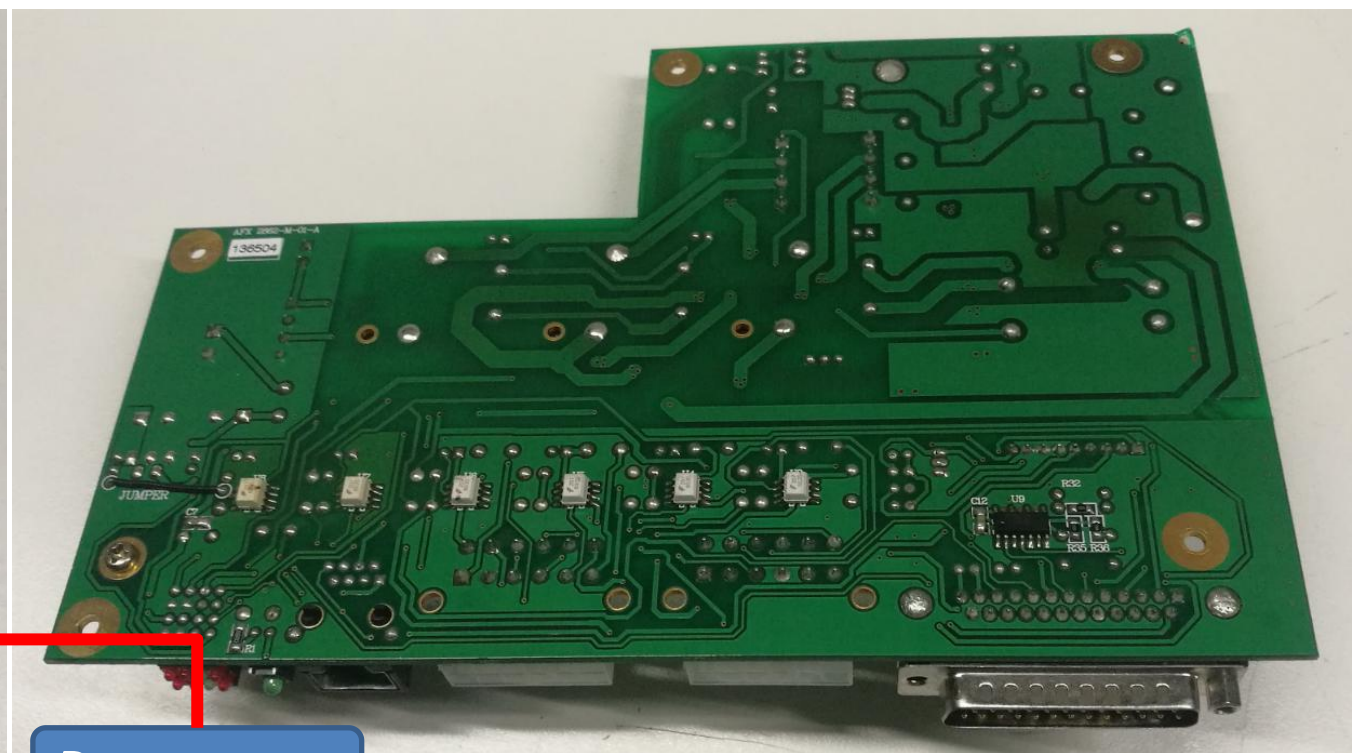
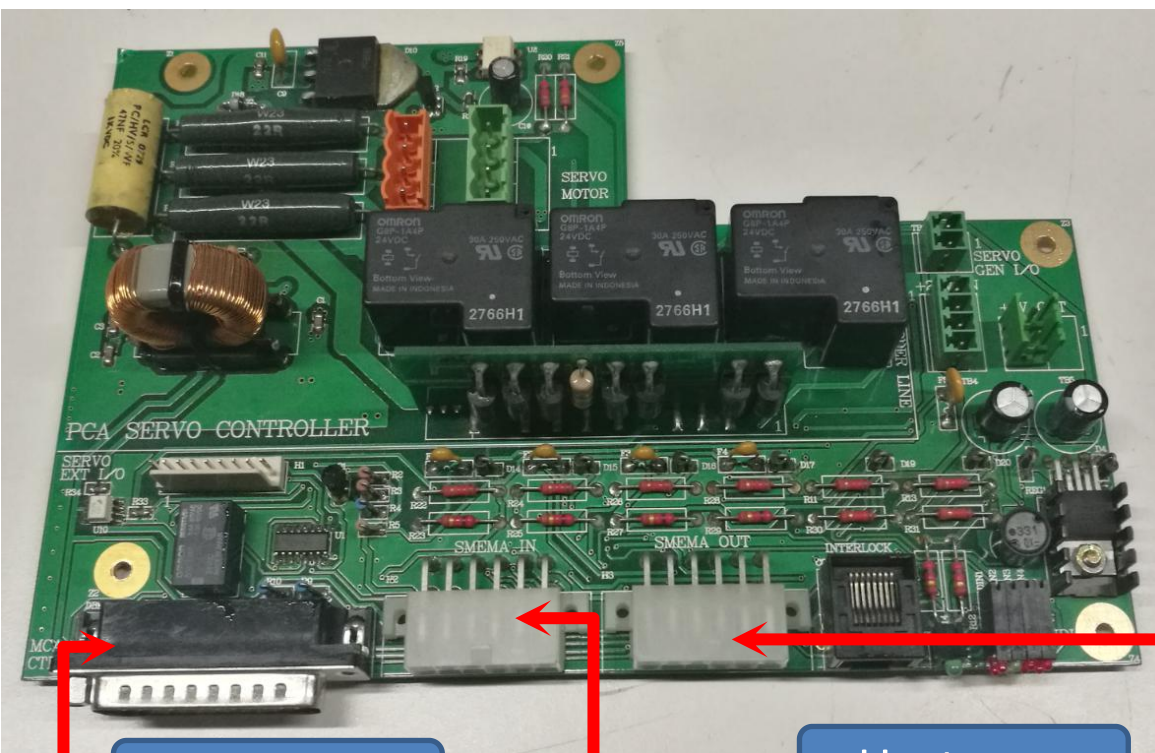


SMEMA Out, Downstream



ABI Servo Interface Board

Automated Board Inspection



To DIO

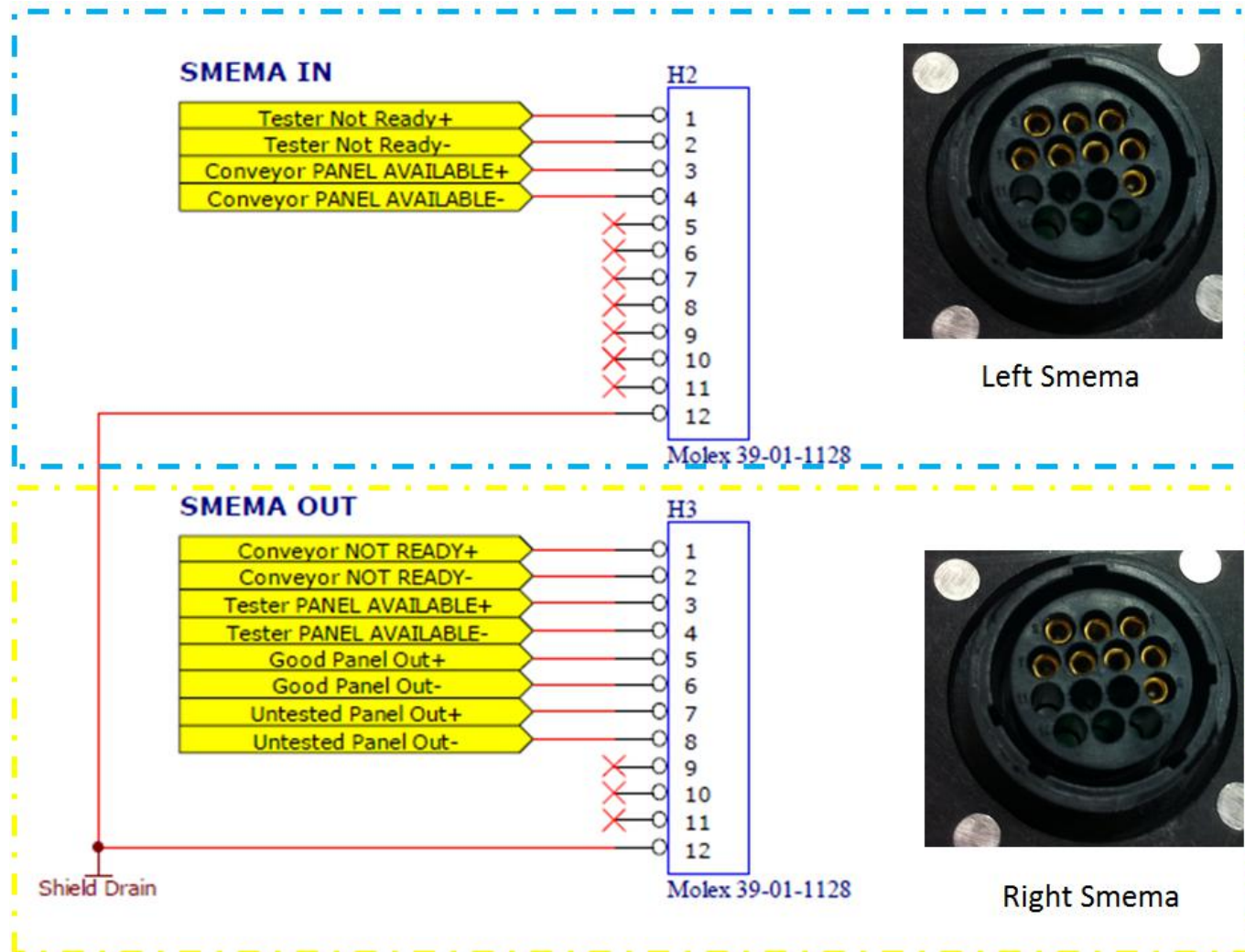
Top

Upstream

Downstream

Bottom

- Designed by using “Optocouplers”, where allow 2 circuits to exchange signal yet remain electrically isolated for switching purpose.
- Not able to measure the output continuity as physically as Optocouplers is not mechanical switch like relay.



ABI Digital I/O tab

Automated Board Inspection

v810 Service - Subsystem Status and Control - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Service Tools Actions Window Help

Display Measurements in: inches (KV: 0.0 uA: 0.0)

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Panel Handling Panel Positioner X-ray Source X-ray Camera Optical Camera Ethernet-Based Light Engine Light Tower SMEMA Interlocks Motion Control Digital IO Connected Processors Hardware Simulation Control

Show Hardware Configuration Description Initialize

Input Bits (Read Only)

- Left Outer Barrier Closed
- Left Panel Clear Sensor Blocked
- Optical Stop Controller Installed
- Interlock Switch ID Bit 4
- Interlock Switch ID Bit 0
- Z-Axis Servo On
- Panel From Upstream System Available
- Left Panel In Place Sensor Extended
- Optical PIP Sensor Reset On
- Z-Axis At High Mag Position 1 On
- X-ray Cameras Connected
- Right Outer Barrier Closed
- Right Panel Clear Sensor Blocked
- Interlock Service Mode Off
- Interlock Switch ID Bit 3
- Inner Barrier Closed
- X-Ray Left Safety Level Sensor On
- Optical Stop Controller Done
- Left Panel In Place Sensor Retracted
- Panel Clamps Opened
- Motorized Height Level Sensor - Low Mag
- X-ray Cameras Powered On
- Left Outer Barrier Open
- Right PIP Optical Stop Sensor Engaged
- Interlock Powered On
- Interlock Switch ID Bit 2
- Inner Barrier Open
- X-Ray Right Safety Level Sensor On
- Optical Stop Controller Error
- Right Panel In Place Sensor Extended
- Z-Axis At Low Mag Position 1 On
- Motorized Height Level Sensor - High Mag
- X-ray Cameras Power In Range
- Right Outer Barrier Open
- Left PIP Optical Stop Sensor Engaged
- Interlock Switch ID Bit 5
- Interlock Switch ID Bit 1
- Z-Axis At Ready Safe Position On
- Downstream System Ready To Accept Panel
- Optical Stop Controller Enabled
- Right Panel In Place Sensor Retracted
- Z-Axis At Low Mag Position 2 On
- Motorized Height Level Sensor - No Alarm
- X-ray Cameras Phase Locked

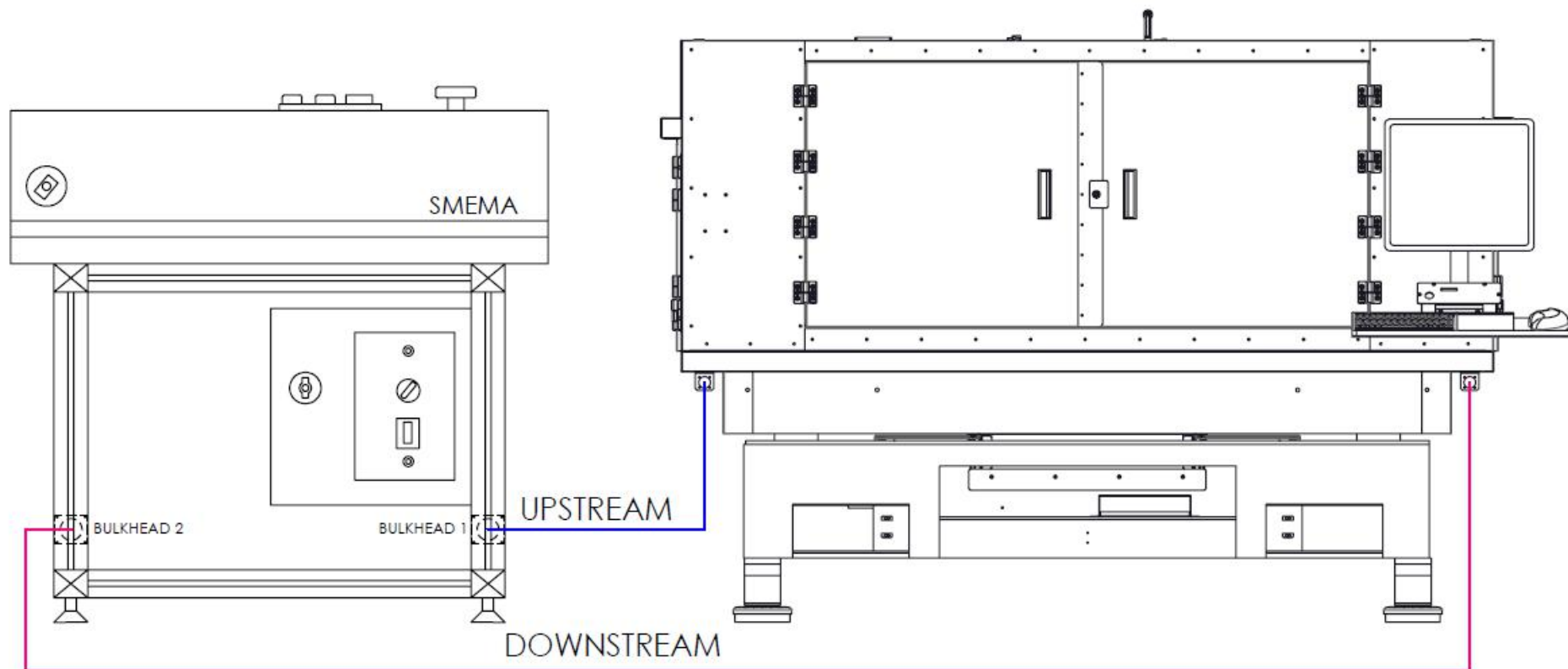
Output Bits (Write)

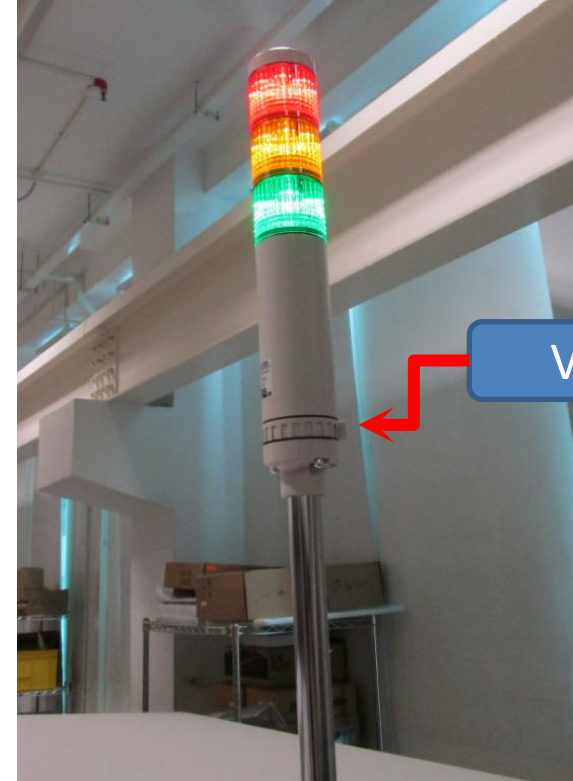
Press button to change state

- Left Outer Barrier Open
- Right Panel In Place Sensor Extended
- Panel Clamps Closed
- Light Stack Yellow Light On
- Z-Axis Position Bit A On
- Panel Test Result Good
- Motorized Height Level Sensor - Start Motion
- X-ray Cameras Synthetic Trigger On Bar
- Right Outer Barrier Open
- Reset Optical PIP Sensor
- Panel Clamps Opened
- Light Stack Green Light On
- Z-Axis Position Bit B On
- Panel From X-ray Tester Not Tested
- Motorized Height Level Sensor - Position Bit0
- Enable Optical Stop Controller
- Stage Belts On
- Inner Barrier Closed
- Light Stack Buzzer On
- X-ray Tester Ready To Accept Panel
- Motorized Height Level Sensor - Reset Alarm
- Motorized Height Level Sensor - Position Bit1
- Left Panel In Place Sensor Extended
- Stage Belts Direction Right To Left
- Light Stack Red Light On
- Z-Axis Enable Signal On
- Panel From X-ray Tester Available
- Motorized Height Level Sensor - Homing
- Motorized Height Level Sensor - Position Bit2

Legend

ON OFF Error



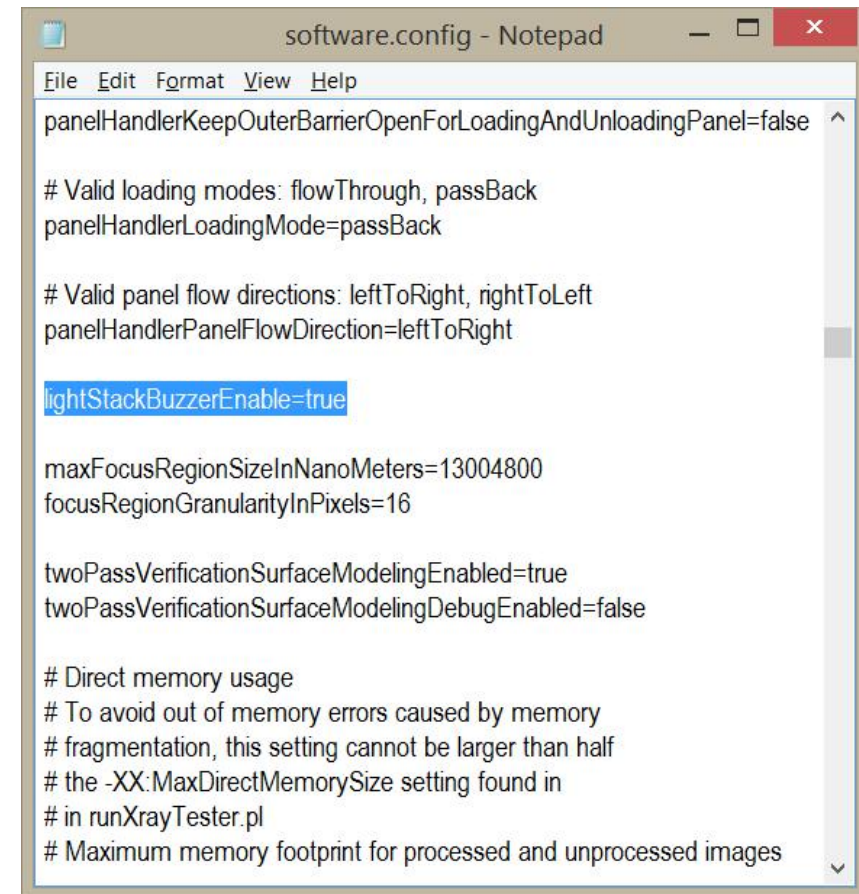
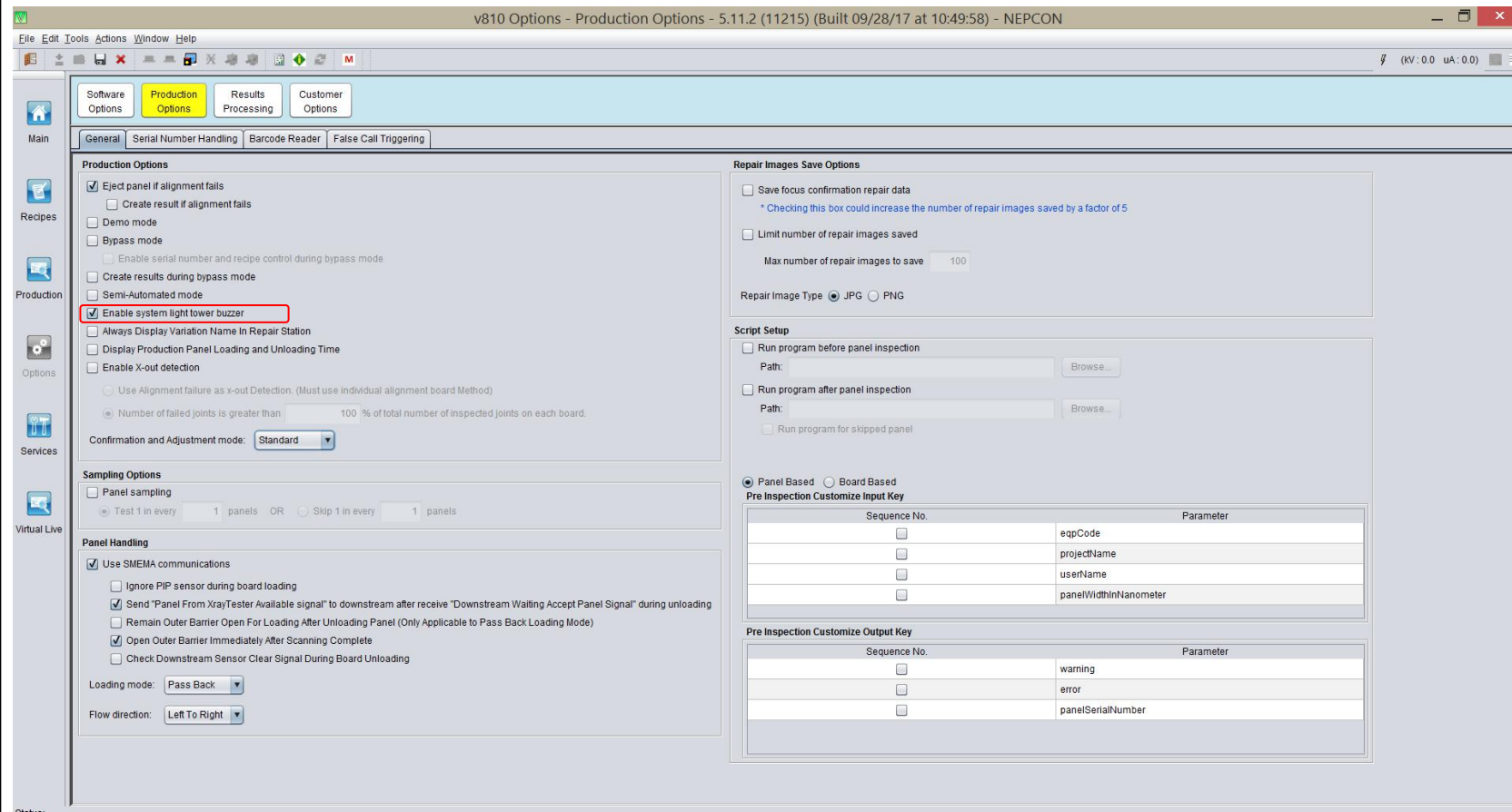


1. Allow user to monitor system condition even have some distance with machine
2. Red stack mean machine error and buzzer will be triggered
3. Yellow stack mean machine in ready condition
4. Green stack mean machine in running condition
5. The Tower Light used is LED type, no bulb change needed



S2EX																							
7-7	7-6	6-7	6-6	6	7	9-7	9-6	8-7	8-6	5-7	5-6	4-7	4-6	3-7	3-6	10-7	10-6	2-7	2-6	1-7	1-6	0-7	0-6
DIO In - 31	DIO In - 30	DIO Out - 31	DIO Out - 30			DIO In - 47	DIO In - 46	DIO In - 39	DIO In - 38	DIO In - 23	DIO In - 22	DIO In - 15	DIO In - 14	DIO In - 7	DIO In - 6	DIO Out - 39	DIO Out - 38	DIO Out - 23	DIO Out - 22	DIO Out - 15	DIO Out - 14	DIO Out - 7	DIO Out - 6
				5V	5V			Optical PIP ErrorSignal	Optical PIP Done Signal	SMEMA Conveyor Panel Available	SMEMA Conveyor Not Ready	Interlock Service Mode On	Interlock Powered On	L.OP-PIP Sensor Engaged	R.OP-PIP Sensor Engaged		Reset Optical PIP Sensor	SHEMA Untested Panel Out	SHEMA Good Panel Out	Z Motor Enable		Innner Barrier Control Valve	Bottom Clamp Up
7-5	7-4	6-5	6-4	4	5	9-5	9-4	8-5	8-4	5-5	5-4	4-5	4-4	3-5	3-4	10-5	10-4	2-5	2-4	1-5	1-4	0-5	0-4
DIO In - 29	DIO In - 28	DIO Out - 29	DIO Out - 28			DIO In - 45	DIO In - 44	DIO In - 37	DIO In - 36	DIO In - 21	DIO In - 20	DIO In - 13	DIO In - 12	DIO In - 5	DIO In - 4	DIO Out - 37	DIO Out - 36	DIO Out - 21	DIO Out - 20	DIO Out - 13	DIO Out - 12	DIO Out - 5	DIO Out - 4
				0V	0V	FH-Motor Alarm	FH-Axis Position Mag #2	Optical PIP Installed	Clamper Open	Right Filter Height Sensor	Left Filter Height Sensor	Interlock Logic Bit #5	Interlock Logic Bit #4	R.Panel Clear Sensor Clear	L.Panel Clear Sensor Clear	FH-Motor Bit #2	FH-Motor Bit #1	SMEMA Tester Panel Available	SMEMA Tester Not Ready			Belt Motor L/R	Belt Motor On/Off
7-3	7-2	6-3	6-2	2	3	9-3	9-2	8-3	8-2	5-3	5-2	4-3	4-2	3-3	3-2	10-3	10-2	2-3	2-2	1-3	1-2	0-3	0-2
DIO In - 27	DIO In - 26	DIO Out - 27	DIO Out - 26			DIO In - 43	DIO In - 42	DIO In - 35	DIO In - 34	DIO In - 19	DIO In - 18	DIO In - 11	DIO In - 10	DIO In - 3	DIO In - 2	DIO Out - 35	DIO Out - 34	DIO Out - 19	DIO Out - 18	DIO Out - 11	DIO Out - 10	DIO Out - 3	DIO Out - 2
Cameras Controller Phase Locked	Cameras Not Using Excessive Current			GND	GND	FH-Axis Position Mag #1	Z-Axis Position Mag #3	Right OSES (PIP) (Retract)	Right OSES (PIP) (Extend)	Z-Motor Output (+)	Z-Axis Position Loading	Interlock Logic Bit #3	Interlock Logic Bit #2	Right Outer Barrier Open	Left Outer Barrier Open	FH-Motor Bit #0	FH-Motor Start	Z-Motor A+	Z-Motor B+	Light Tower Buzzer	Light Tower Green	Right OSES (PIP) Extended	Left OSES (PIP) Extended
7-1	7-0	6-1	6-0	1	0	9-1	9-0	8-1	8-0	5-1	5-0	4-1	4-0	3-1	3-0	10-1	10-0	2-1	2-0	1-1	1-0	0-1	0-0
DIO In - 25	DIO In - 24	DIO Out - 25	DIO Out - 24			DIO In - 41	DIO In - 40	DIO In - 33	DIO In - 32	DIO In - 17	DIO In - 16	DIO In - 9	DIO In - 8	DIO In - 1	DIO In - 0	DIO Out - 33	DIO Out - 32	DIO Out - 17	DIO Out - 16	DIO Out - 9	DIO Out - 8	DIO Out - 1	DIO Out - 0
Camera Control Power On	Camera Control Cables Connected	Cameras Control Spare	Cameras Synth Clock			Z-Axis Position Mag #2	Z-Axis Position Mag #1	Left OSES (PIP) (Retract)	Left OSES (PIP) (Extend)	Inner Barrier Open	Inner Barrier Closed	Interlock Logic Bit #1	Interlock Logic Bit #0	Right Outer Barrier Closed	Left Outer Barrier Closed	FH-Motor Home	FH-Motor Alarm Reset	Enable Optical PIP Controller	Panel Clamps Open	Light Tower Yellow	Light Tower Red	Right Outer Barrier Control Valve	Left Outer Barrier Control Valve
Slice 7 - Input		Slice 6 - Output		Field Power Distributor		Slice 9 - Input		Slice 8 - Input		Slice 5 - Input		Slice 4 - Input		Slice 3 - Input		Slice 10 - Output		Slice 2 - Output		Slice 1 - Output		Slice 0 - Output	
																						MES Slice IO Controller	

➤ Digital Input/Output Slide and Motion Scope Traces



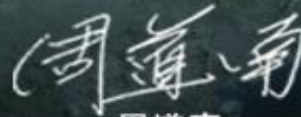
- Buzzer can be deactivated with set key “lightStackBuzzerEnable” as “False” in Software.config or Option> Production Option tab



WEE KAH KHIM
Vice President & General Manager



GARY LEONG
Sales Director



周道南
Sales General Manager
(China & Taiwan)



SEOW ZI YANG
Business Development
Manager



FAHMI HELMI
Business Development
Manager